

occurrences using the simple criterion of predictability, which makes them all functionally equal. Probability can be degrees of belief, what one uses to make a bet, or something more physical associated with true randomness (called “ontic,” on which later). To paraphrase Gerd Gigerenzer, a “50 percent chance of rain tomorrow” in London might mean that it will rain half the day, while in Germany it will mean that half the experts think it will rain, and (I am adding), in Brooklyn, that the betting market at the bar is such that one would pay 50 cents to get a dollar if it rains.

For scientists, the treatment is the same. We use the same equation to describe a probability distribution, regardless of whether the probability is a degree of belief or something designed by Zeus, who, we believe, calls the shots. For us probabilists (persons who work with probability in a scientific context), the probability of an event, however it may be defined, is, simply, a weight between 0 and 1, called the measure of the set concerned. Giving different names and symbols would be distracting and would prevent the transfer of analytical results from one domain to another.

For a philosopher, it is altogether another matter. I had two lunches with the (analytical) philosopher Paul Boghossian, three years apart, one upon the completion of the first edition of *The Black Swan*, the second upon the completion of this essay. During the first conversation he said that, from a philosophical point of view, it is a mistake to conflate probability as a measure of someone’s rational degree of belief with probability as a property of events in the world. To me, this implied that we should not use the same mathematical language, say, the same symbol, p , and write down the same equation for the different types of probabilities. I spent three years wondering if he was right or wrong, whether this was a *good redundancy*. Then I had lunch with him again, though in a better (and even more friendly) restaurant.

He alerted me to a phrase philosophers use: “distinction without a difference.” Then I realized the following: that there are distinctions philosophers use that make sense philosophically, but do not seem to make sense in practice, but that may be necessary if you go deeper into the idea, and may make sense in practice under a change of environment.

For consider the opposite: differences without a distinction. They can be brutally misleading. People use the same term, *measuring*, for measuring a table using a ruler, and for measuring risk—when the second is a forecast, or something of the sort. And the word *measuring* conveys an illusion of knowledge that can be severely distorting: we will see that we are psychologically very vulnerable to terms used and how